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QEM-9: Hyperstacked Mirror Lattices under the Opposition Principle (Public v2.1)

QEM-9□□□□□□□□□□□□□□□□ v2.1□

Author: Youfu Wang (□□□) | ORCID: 0009-0005-5998-6024 | Date: 2025-11-07 | License: CC BY 4.0

Abstract / □□

This paper extends the QEM framework from 3D to d-dimensional hyperstacked mirror lattices under a Σ/Δ Opposition Principle. Admissible transport and stabilization reside in the odd (Δ) sector, yielding sector-resolved bounds and coherence scaling $\xi_d \approx \xi_1/\sqrt{d}$ under bounded Σc_k^2. A design-for-opposition SOP is summarized for scalable architectures.

□□□ QEM □ 3D □□□ d □□□□□□□ Σ/Δ □□□□□□□□ Δ□□□□□□□□□□□□□□□□ Σc_k^2 □□□□□□□□□□ $\xi_d \approx \xi_1/\sqrt{d}$□□□□□□□□□□□□□ SOP□

1. Framework

Opposition operators $O_k \in \{\pm 1\}$ per axis; global $O^{(d)} = \otimes_k O_k$. Transport/stability computed in odd sector using product-Laplacian spectra.

□□□□□□□□□ O_k□□□□□ Δ □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

2. Bounds (Public) / □□□□□

Propagation: $v_{max}^{(d)} \approx \min(v_{local}, (1/\pi)\sqrt{(a_{\parallel}^2 \Omega_0^2 + \Sigma\, a_k^2 c_k^2 \lambda_{max,k}^{odd})})$.

Stability island & delay: $\tau_c(d) < 0.2 \cdot \min_k \{\xi_k / c_k\}$.

Coherence: $\xi_d \approx \xi_1 / \sqrt{d}$ (if $\Sigma\, c_k^2$ bounded).

3. SOP (Public) / □□ SOP

Bipartite layout; loop-parity (even Σ per closed loop); π-inverters on odd-Σ paths; sector-selective validation (parity-echo, basis-flip).

□□□□□□□□ Σ □□□□□□□ π □□□□□□□□□□□□□□□□□□□□□□□□

4. Disclosure Note / [illegible]

This public version omits device-level parameter ranges, detailed latency budgets, and fabrication-tied values. Implementation notes are kept in a confidential companion document for R&D use.

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